

PAC Learning

Kyle Macadaeg

Definition

Definition 1.3 – PAC Learning: Take II. For $n \geq 1$, let C_n be a concept class over instance space X_n and let $C = \bigcup_{n \geq 1} C_n$ and $X = \bigcup_{n \geq 1} X_n$. We say that C is PAC (take II) learnable if there exists a learning algorithm L that satisfies the following: for every $n \in \mathbb{N}$, for every concept $c \in C_n$, for every distribution D over X_n , for every $0 < \epsilon < 1/2$ and $0 < \delta < 1/2$, if L is given access to $\text{EX}(c, D)$ and inputs n , $\text{size}(c)$, ϵ and δ , L outputs $h \in C_n$ that with probability at least $1 - \delta$ satisfies $\text{err}(h) \leq \epsilon$. The probability is over the random examples drawn from $\text{EX}(c, D)$ as well as any internal randomisation of L . The number of calls made to $\text{EX}(c, D)$ (sample complexity) must be bounded by a polynomial in n , $\text{size}(c)$, $\frac{1}{\epsilon}$ and $\frac{1}{\delta}$.

We further say that C is efficiently PAC (take II) learnable if the running time of L is polynomial in n , $\text{size}(c)$, $1/\epsilon$ and $1/\delta$.

Okay, I'm joking. (Kinda.)

I think we're already familiar with the definition at this point!

Let me tell you a story...

Okay, I'm joking. (Kinda.)

Two people who know something might think about that same thing in two different ways.

Let's try to view PAC Learning a bit differently today.

PAC Learning

Kyle Macadaeg

PAC Learning

is inescapable

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My True Goal Today

By “inescapable”, I mean that it can show up in surprising places.

My goal is to:

- Convince everyone that PAC Learning is inescapable.
- Touch on what makes it so inescapable.

Revisiting the Definition

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We further say that C is efficiently PAC (take II) learnable if the running time of L is polynomial in n , $\text{size}(c)$, $1/\epsilon$ and $1/\delta$.

δ parameter: We may not always produce a good hypothesis

ϵ parameter: Good hypotheses may not necessarily be always correct
“Efficiently”: There needs to be some level of “elegance” to our method.

Revisiting the Definition

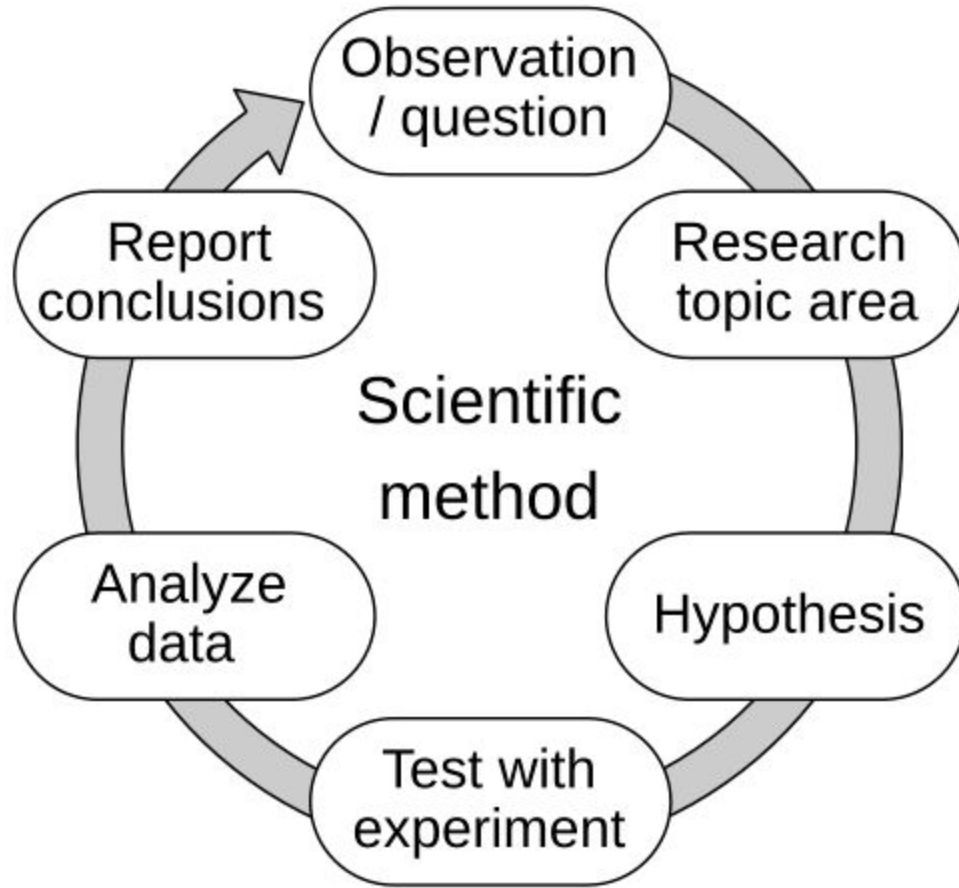
Definition 1.3 – PAC Learning: Take II. For $n \geq 1$, let C_n be a concept class over instance space X_n and let $C = \bigcup_{n \geq 1} C_n$ and $X = \bigcup_{n \geq 1} X_n$. We say that C is PAC (take II) learnable if there exists a learning algorithm L that satisfies the following: for every $n \in \mathbb{N}$, for every concept $c \in C_n$, for every distribution D over X_n , for every $0 < \epsilon < 1/2$ and $0 < \delta < 1/2$, if L is given access to $\text{EX}(c, D)$ and inputs n , $\text{size}(c)$, ϵ and δ , L outputs $h \in C_n$ that with probability at least $1 - \delta$ satisfies $\text{err}(h) \leq \epsilon$. The probability is over the random examples drawn from $\text{EX}(c, D)$ as well as any internal randomisation of L . The number of calls made to $\text{EX}(c, D)$ (sample complexity) must be bounded by a polynomial in n , $\text{size}(c)$, $\frac{1}{\epsilon}$ and $\frac{1}{\delta}$.

We further say that C is efficiently PAC (take II) learnable if the running time of L is polynomial in n , $\text{size}(c)$, $1/\epsilon$ and $1/\delta$.

In total, this makes it a pretty good model for the process of learning in general.

This shouldn't be a surprise; of course machine learning is meant to emulate human learning in some capacity.

Part 1: Science and Epistemology



Use data to inform a hypothesis.

Seems familiar, doesn't it?

Inductivism

- Induction (not to be confused with mathematical induction) is reasoning about future observations based on the past ones.
- For example, if you see 500 ravens that are black, you reason that the 501st raven will also be black.
- The PAC Learning setting is inherently inductivist; we use data to inform our selection of a hypothesis.
- Think of the PAC Learning setting *not* as the application of inductivism to machine learning, but a formalization of inductivism.

Challenges to Inductivism

Point 1: Epistemologically, why should we believe that the 501st raven is black?

Point δ : We can produce hypotheses of poor explanatory power.

- Light behaves as a wave, and thus space must be filled with some sort of “luminiferous aether” through which light can propagate.

Point ϵ : Our hypotheses are never perfect.

- Newtonian mechanics is good... as long as you stay macroscopic.
- We have yet to produce a “theory of everything”, a physical theory that can explain both relativity and quantum mechanics.

Popperian Philosophy

- Inductivism doesn't actually produce knowledge.
 - The scientific method can only produce and disprove hypotheses, never prove them.
-
- By the mere existence of (δ, ϵ) , we never claimed to produce knowledge anyway; just our best guess.
 - Nevertheless, inductivism seems to be a quite powerful tool.

The 501st Raven

“All ravens are black” is a much simpler hypothesis than “all ravens are black except for the 501st raven”.

This seems to address the epistemological point, but creates a few points of its own.

- What do we mean by simpler?
- Why should simpler hypotheses be preferred?
- How much data is needed to create a strong hypothesis?

What do we mean by simpler?

- Not something that we can entirely address.
- In the PAC learning setting, the size of the concept is the obvious choice (for example, Kolmogorov complexity).
- We run into the obvious issue of how we translate hypotheses into strings.

Nelson Goodman's "New Riddle of Induction": Are the words "grue/bleen" meaning "green/blue before January 1st, 2030 and blue/green afterwards" really simpler than "green/blue"? It seems obvious on the surface, but we can imagine that "grue" and "bleen" are the more "primitive" concepts, with "green/blue" meaning "grue/bleen before January 1st, 2030 and bleen/grue afterwards".

Why should simpler hypotheses be preferred?

- This is something that CoLT can actually provide an answer for.
- In CoLT, we have Occam learners: a consistent learner that outputs a hypothesis whose size is polynomially bounded.
- We have proven that an Occam learner can be adapted into a PAC learner by simply drawing enough samples.
- This gives us a reason to favor simpler hypotheses: as long as they are consistent with our data, they generalize well.
- Think of CoLT's Occam's Razor not as an application of Occam's Razor to machine learning, but as a formalization of Occam's Razor.
- Our theorem provides credence to Occam's Razor, and thus inductivism.

How much data is needed to create a strong hypothesis?

Polynomially many, since an efficient Occam learner is an efficient PAC one.

Part 2: Language Acquisition and Psychology

Some Caveats

We're going to talk about the process of acquiring language. In comparison to inductivism, this is a much harder thing to try to formalize, and in doing so we make a few assumptions.

1. The input data has no noise, i.e. a child will only ever receive correctly formed sentences of their language.
2. The way in which all children learn language can be represented by a singular algorithm.
3. That PAC is the right setting here.

Is PAC Learning the right setting here?

The requirement of PAC learning to work for any setting of (δ, ϵ) might be a bit too strict of a requirement.

At the same time, for us to talk about PAC learning, we need a concept class that is known to the learner (in our instance, this will be different types of languages).

In (partial) defense:

- Point δ : Some people struggle with language more than others.
- Point ϵ : Even those who know their language well are prone to small grammatical mistakes or other minutiae.
- Point 3: Efficiency is still needed here, since children learn languages fast.

Some Prerequisites

Noam Chomsky was interested in what linguistics reveals about the human mind.

He posited about the existence of a “Universal Grammar”, some sort of predisposition or bias that humans have when they learn languages.

He also came up with the “Chomsky hierarchy” that describe grammars.

Grammar ↕	Languages ↕	Recognizing automaton ↕	Production rules (constraints) ^[a] ↕	Examples ^{[5][6]} ↕
Type-3	Regular	Finite-state automaton	$A \rightarrow a$ $A \rightarrow aB$ (right regular) or $A \rightarrow a$ $A \rightarrow Ba$ (left regular)	$L = \{a^n \mid n > 0\}$
Type-2	Context-free	Non-deterministic pushdown automaton	$A \rightarrow \alpha$	$L = \{a^n b^n \mid n > 0\}$
Type-1	Context-sensitive	Linear-bounded non-deterministic Turing machine	$\alpha A \beta \rightarrow \alpha \gamma \beta$	$L = \{a^n b^n c^n \mid n > 0\}$
Type-0	Recursively enumerable	Turing machine	$\gamma \rightarrow \alpha$ (γ non-empty)	$L = \{w \mid w \text{ describes a terminating Turing machine}\}$

Some Prerequisites

Each type contains all of the languages of the preceding types.

More concerningly, any fixed set of strings are a subset of infinitely many languages (regardless of type).

This is what motivates the “Universal Grammar” idea: how is it that we all can agree on what an English sentence is, even when there are infinitely many languages that could contain all of the sentences we will ever hear?

If we can show that language learning is not effectively PAC, it lends credence to the fact that we have some sort of innate predisposition to certain languages.

Are any of the types PAC learnable?

Let's consider finite languages. Learning a language is then learning a finite subset of words from the set of all words. We've seen before that this has infinite VC dimension.

Let's consider finite languages with words up to size n . There are still exponentially many languages, and the VC dimension grows super-polynomially.

All finite languages are regular (Type 3), which means all finite languages are also Type 2, 1, and 0. If finite languages cannot be learned, certainly these can't either.

Guess we're done!

An Objection

But wait, what about membership queries?

Learning a language is a two-way process. Children don't just listen; they speak too.

In our setting, a membership query would be akin to a child saying a complete sentence and being told whether or not it is correctly formed. This is a reasonable assumption.

An Objection

In fact, membership queries do expand our powers somewhat. With membership queries, you can learn DFAs efficiently (a proof of this is in the Kanade book). Since DFAs recognize regular languages, we can learn regular languages with membership queries.

However, regular languages are Type 3. Most linguists believe that natural languages are Type 1, with some arguing they are Type 2. So even still, we're out of luck.

An Objection

We *can* learn a strict subset of the context-free languages which is also a strict superset of the regular languages... if we fix not only a target language but a target grammar for that language and extend the definition of membership queries to allow for queries on non-terminals that are not the starting symbol.

This would be akin to a child asking “Is X a prepositional phrase?” which of course is not how we learn languages (at least for the first time).

Plus, this still doesn't get us to Type 1. It is known that these languages are not PAC+MQ learnable unless P does not equal NP .

Conclusion

PAC learning can show up in surprising places and get surprising results.

Its generality lends itself to being adapted in different fields.

So with this, let's tell one last story.

Kurt Gödel

Kurt Gödel must have been implicitly aware of just how powerful it is to allow for exponential time. On the topic of recognizing theorems from axioms, he writes:

If there actually were a machine with [running time] $\sim Kn$ (or even only with $\sim Kn^2$) [for some constant K independent of n], this would have consequences of the greatest magnitude. That is to say, it would clearly indicate that, despite the unsolvability of the Entscheidungsproblem, the mental effort of the mathematician in the case of yes-or-no questions could be completely [added in a footnote: apart from the postulation of axioms] replaced by machines. One would indeed have to simply select an n so large that, if the machine yields no result, there would then also be no reason to think further about the problem.

Kurt Gödel

He implicitly also seems to be cognizant of this fact on the topic of evolution.

I believe that mechanism in biology is a prejudice of our time which will be disproved. In this case, one disproof, in my opinion, will consist in a mathematical theorem to the effect that the formation within geological time of a human body by the laws of physics (or any other laws of similar nature), starting from a random distribution of the elementary particles and the field, is as unlikely as the separation by chance of the atmosphere into its components.

Notice the caveat “within geological time”.

Wouldn't it be nice if we could show that evolution can be done efficiently?

After all...

Point δ : There are many species that have gone extinct.

Point ϵ : Even among members of existent species, some are unable to reproduce.

So it looks like even here, PAC learning is inescapable.

Sources

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